Science Academies' two-week refresher course in Quantum Mechanics (17–31 December 2017)

A brief report

The Department of Physics, in association with the Indian Academy of Sciences, the National Academy of Sciences and the Indian National Science Academy, organised a national-level refresher course in our college from 17–31 December 2017 that drew participants and renowned speakers from across the country. The course targeted physics teachers working at undergraduate colleges and universities. The course was formally titled a 'Refresher course in Quantum Mechanics with a special emphasis on Nuclear and Particle Physics'.

This college, like several others in India, offers condensed matter physics as a field of specialisation in its postgraduate course; several others institutions also offer nuclear physics as a specialisation. Both these fields rely considerably on the ideas of quantum mechanics, a branch of physics that is as hard to grasp as it is beautifully complex. One of the best methods of keeping up with current physics is to have a good understanding of quantum mechanics, which is what motivated our department to host this refresher course. Inviting participants from our institutions along with those from institutions around the country made it an excellent opportunity to learn from and network with others in our field.

It was therefore our plan to start the refresher course with an overview of quantum mechanics at the higher undergraduate level before proceeding to discuss its more advanced ideas, connecting it to special relativity, and finally applying these ideas to improve our understanding of nuclear and particle physics.

The event was inaugurated on the morning of 17 December with Prof. K. Chidananda Gowda, the former vice-chancellor of Kuvempu University, present as the chief guest alongside the guest of honour Prof. R. Srinivasan, retired director of the UGC–DAE Consortium for Scientific Research. The rector/manager of our college, Rev. Fr. Bernard Prakash Barnis, presided over the inaugural session. Also present, besides the participants, were the course director Prof. H.S. Mani of the Chennai Mathematical Institute and the co-ordinator Prof. D. Revannasiddaiah.

Forty-five participants from all over the country were present during the course with close to twenty from outside Karnataka. Professors from Rajasthan, Bihar, Arunachal Pradesh and West Bengal were among the participants in this course besides those from Kerala and Andhra Pradesh. There were almost an equal number of participants from within our state coming from various districts including Hubli–Dharwad, Gulbarga and Bangalore along with participants from local universities and colleges. Besides teachers highly motivated postgraduate students and research scholars were carefully selected to participate in this course. This included students from our own college and the postgraduate centres of the University of Mysore. Candidates from outside our city were put up in our college hostel and at the Organisation for the Development of People (ODP) beside our campus.

Each day of the fortnight-long course was broken up into four ninety-minute sessions for lectures and tutorials. The course was conducted in the physics lecture hall on the first floor of the postgraduate block. Prof. Mani started his lectures with an overview of tensors and special relativity, alternating with Prof. K.S. Mallesh from the University of Mysore who gave an overview of quantum mechanics from the perspective of its applications in nuclear physics. Most afternoon

sessions were spent solving problems based on problem sets given out by the two speakers. By the end of the first week they had laid a solid foundation on the fundamental ideas required for discussing particle physics and had set the stage for more advanced talks.

The first phase of the course was concluded with four special lectures by Prof. V. Ravishankar of IIT Delhi who spoke on quantum chromodynamics focussing on the quark-gluon plasma. In addition to this, although Prof. Thyagarajan gave an independent lecture on turbulence outside the regular course schedule, stretching beyond 6 pm, all participants showed great interest in attending it and indulged themselves in discussions after the fact.

The next phase of the course was taken up by Prof. M.V.N. Murthy of the Institute of Mathematical Sciences who spoke on particle physics. Over the remainder of the course he covered various ideas related to the quark model with remarkable clarity ending with a mathematically precise theory on the structure of quarks inside subatomic particles that made predictions to be tested through experiments.

Also present was experimental physicist Prof. Vandana Nanal from the Tata Institute of Fundamental Research who addressed various past and current experimental approaches towards studying the nucleus and nuclear physics in general. She began with an overview of the nucleus and its discovery and went on to discuss, among other ideas, techniques used to measure the radii of nuclei.

In-between sessions the corridors of the postgraduate block were filled with participants interacting with resource persons. It was an immense pleasure for us to witness so many young physicists making use of this opportunity provided by our college to make a difference to, and help improve, their careers.

The course was wrapped up by Prof. D. Indumati of the Institute of Mathematical Sciences who gave a quick summary of most topics discussed so far in the context of the Standard Model of particle physics, which formed the crux of her lectures. She described the Standard Model in detail and delved into the Higgs mechanism and weak interactions, two important discussions in contemporary particle physics.

A brief session by Mr Bhaskar of TNQ was held on the penultimate day where he introduced Author Café a collaborative online writing tool that many participants and resource persons took interest in.

Every candidate was given four books—selected by the resource persons and related to the course topics—worth approximately Rs 1,500 in total. Additionally, towards the end of the course it was decided that select course material provided by the speakers would be made available to candidates via e-mail and on our event page online. A feedback form was provided to participants in which everyone indicated that they were extremely pleased with the organisation, content and structure of the programme and several candidates acknowledging the college for its role in helping them deepen their interests in physics both from the perspective of a teacher and a researcher.

Prof. M.V.N. Murthy was present as the chief guest during the valediction with our college principal, Prof. T. Ruth Shanthakumari presiding. The rector/manager, Rev. Fr. Bernard Prakash Barnis, was the guest of honour and distributed certificates of participation before formally ending the course with Prof. D. Revannasiddaiah, the head of the postgraduate department of physics, acknowledging those who helped make this course a success.

Like the national-level refresher course in experimental physics conducted in February 2017 by our department, this event too was a landmark for our college. It worked with the three Science Academies of India and dealt with important topics of modern-day physics that would assist in the careers of participating students and professors alike. As India heads into 2018 with several international collaborations in nuclear and particle physics scheduled to begin this year, this refresher course is exemplary of the passionate teaching and of the research knowledge on par with the frontiers of contemporary physics in our department and, by extension, in our college.

Faculty,
UG and PG Departments of Physics